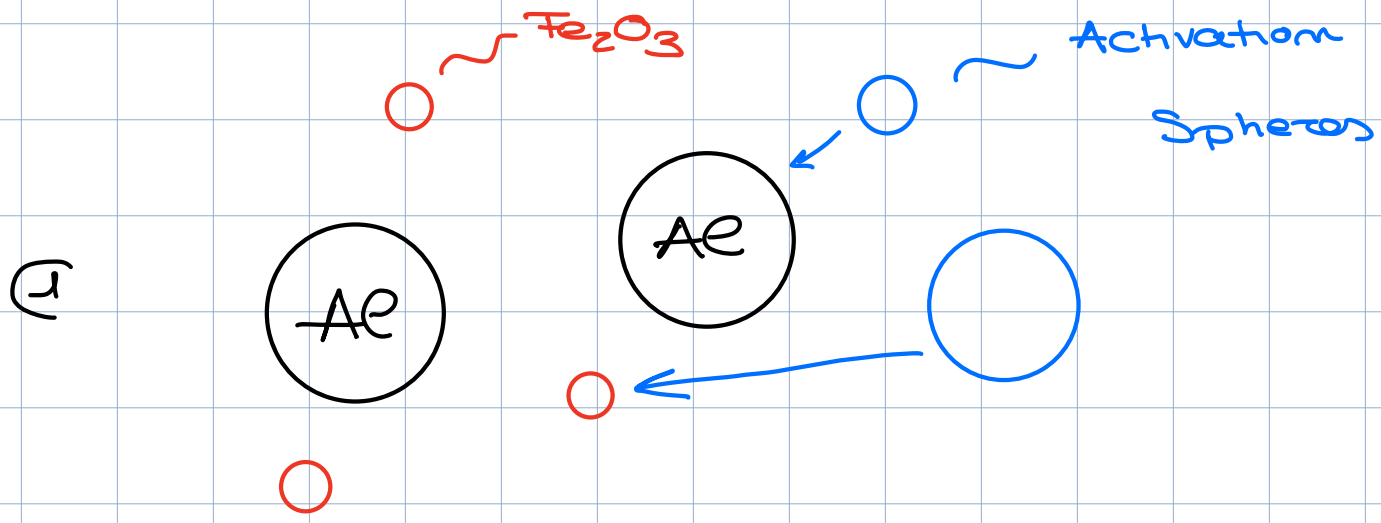
⓪ Starting point:

i) Particles are in the jar, with spheres

ii) Jar rotates



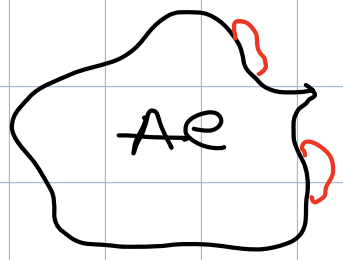
① In the "activation jar", spheres impact on the powder particles

=> (i) particle morphology changes

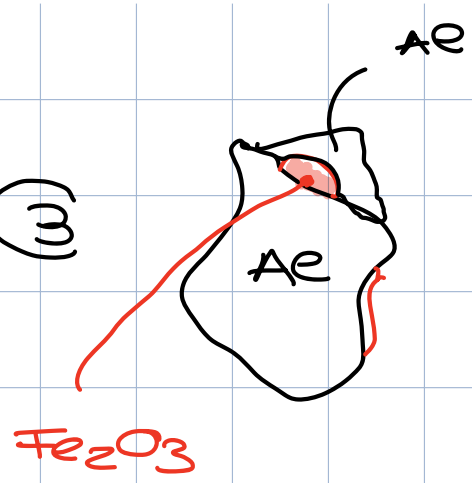
(ii) particles are "packed" by the impacts

(iii) depending on operating conditions, particles are welded (size ↑) or grinded (size ↓) we do not focus on "what operating condition does what"

②



③



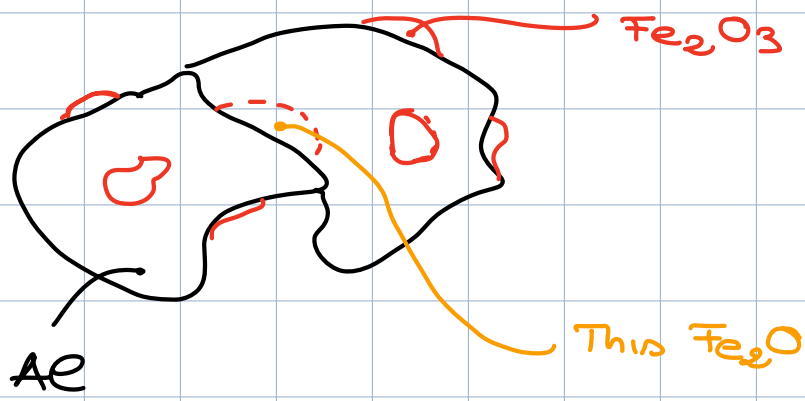
② & ③ Spherical shape of Al is lost due to the mechanical stresses & impacts.

For the same reasons, Fe_2O_3 (in general, "the other powder") impacts on Al $\Rightarrow$ Al & Fe_2O_3 are in close proximity!

When further collisions occur, Al can "include" Fe_2O_3

$\Rightarrow$ composite powder

④



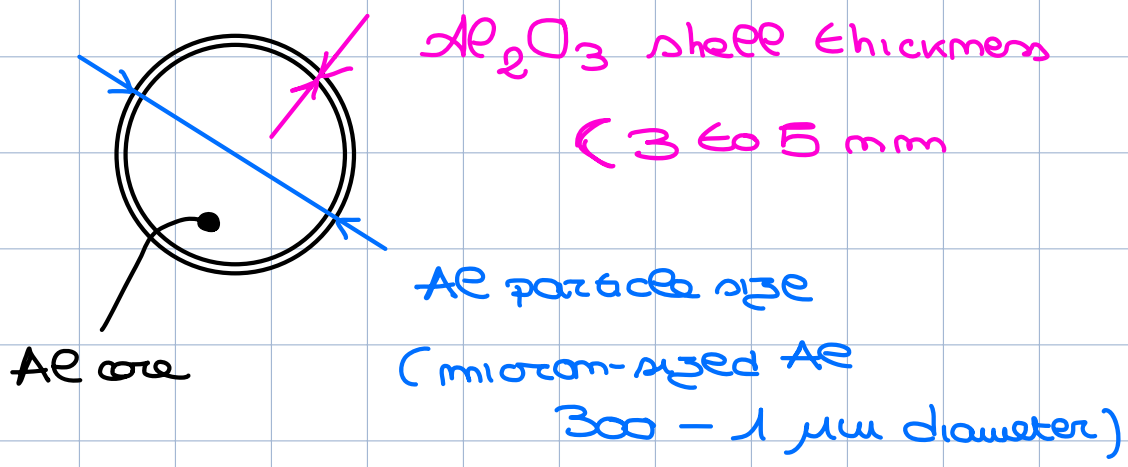
This Fe_2O_3 is "inside the composite particle"

④ $\Rightarrow$ in the end, some particle of Fe_2O_3 is "inside"

⇒ Mechanical activation reduces the diffusion distance!

PS:

- actual Al particles feature a core-shell structure (Al core surrounded by Al_2O_3 shell; shell thickness is 3–5 nm ⇒ negligible for micron-sized particles)



Specific Surface Area (SSA) & Particle Radius (R)

$$SSA = [m^2/g]$$

Particle surface

$$= \frac{4\pi R^2}{\frac{4}{3}\pi R^3 \rho} \Rightarrow SSA \sim 1/R$$

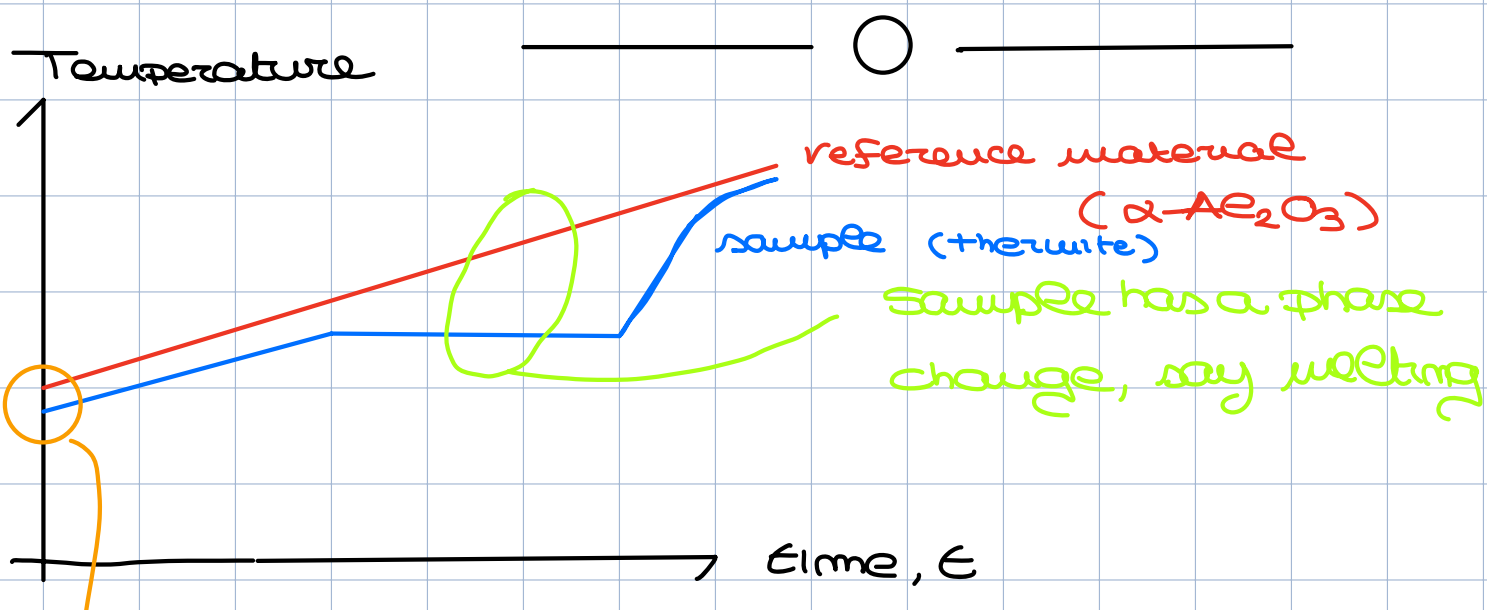
Particle mass

(volume, if "ρ" is neglected)

SSA ↑ as R ↓

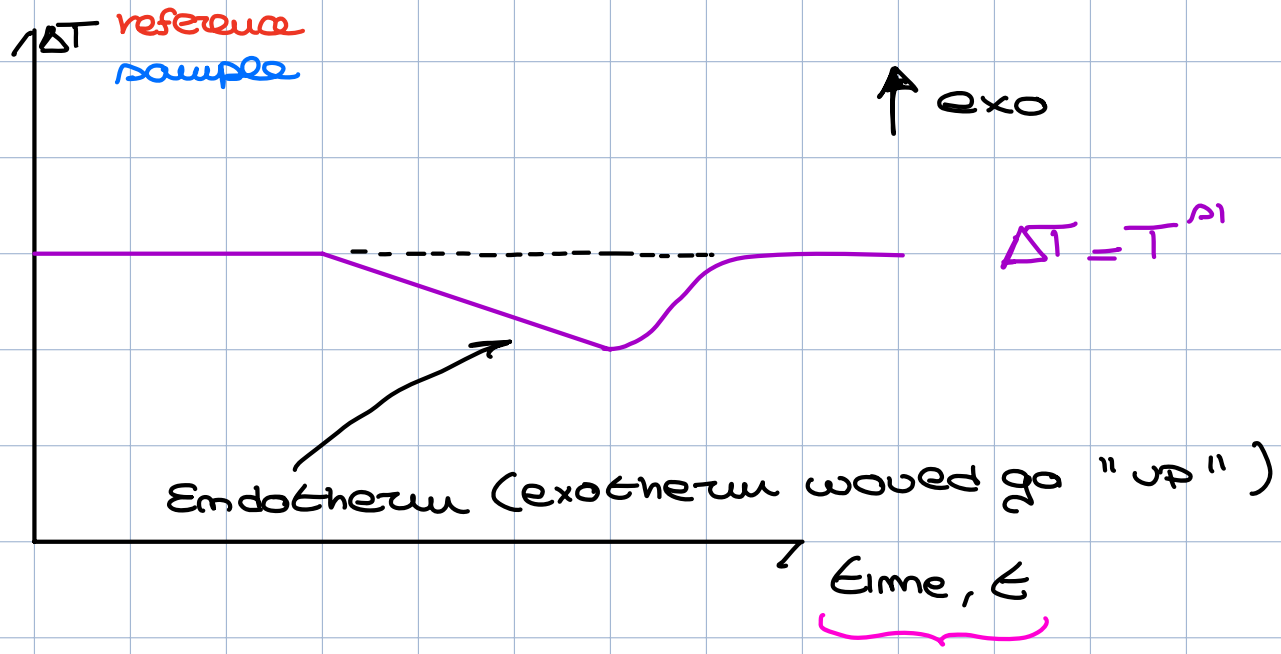
(Remember what we said on "the increasing relative weight of surface as volume decreases)

→ Finer particles (say, nanoscale Al) show higher SSA (and faster reactivity)



This shift is just for representative purposes, actually here $\Delta T^{\text{reference}}_{\text{sample}} = 0$

make the difference between the red and the green lines



Heating rate is steady

$\Rightarrow$ time $\equiv$ Temperature